1. a) heating load = 5 KW

Motor load = 50 KVA at 0.6 pf (lag) = 30 KW + j 40 KVAR

so, Total load = 35 KW + j 40 KVAR

$$S = V I^*$$

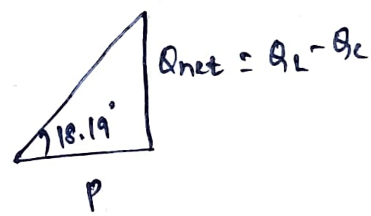
$$I = \left(\frac{S}{V}\right)^* = \left[\left(\frac{35 + j40}{240}\right) \times 10^3\right]^*$$

$$\boxed{I = 221.46 \angle -48.81^\circ \text{ A}}$$

(b) plant power factor = $\cos(48.81)_{lag}$ = 0.66 lag

(c) Now, improved power factor = 0.95 lag

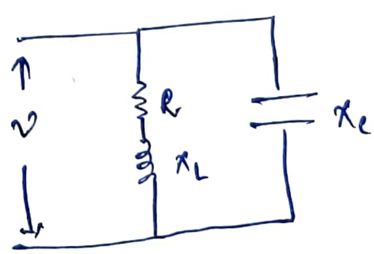
i.e. $\phi = \cos^{-1}(0.95) = 18.19^\circ$



$$\text{so, } \tan(18.19^\circ) = \frac{Q_L - Q_c}{P}$$

$$0.329 = \frac{Q_L - Q_c}{35 \text{ KW}}$$

$$0.329 = \frac{40 - Q_c}{35 \text{ KW}}$$



$$\boxed{Q_c = 28.5 \text{ KVAR}}$$

$$\text{so, } \frac{V^2}{X_c} = Q_c$$

On calculation $\boxed{C = 1.57 \text{ mF}}$ Put $\boxed{\omega = 100\pi \text{ rad/s}}$

(d) so, Net power Demand = $P_{net} + jQ_{net}$

$= 35\ kW + j(40-28.5)\ kVAR$

$= 35\ kW + j\,11.5\ kVAR$

$$I = \left(\frac{P}{V}\right)^* = \left[\left(\frac{35 + j\,11.5}{240}\right) \times 1000\right]^*$$

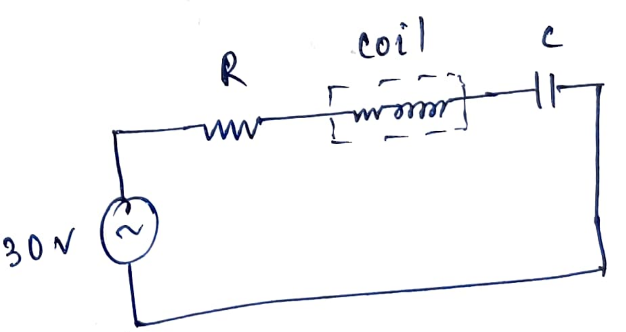
$$\boxed{I = 153.5 \angle -18.18^\circ\ A}$$

(e) yes, addition of capacitor in parallel to R-L load improves the power factor and thus reduces the overall demand of current.

(2) Given $C = 0.5\ \mu F$

coil $\longrightarrow$ $R = 40\ \Omega$, $L = 0.32\ H$

$R = 20\ \Omega$



$$\omega_0 = \frac{1}{\sqrt{LC}} = 2500\ rad/s$$

$$f_0 = \frac{\omega_0}{2\pi} = 397.89\ Hz$$

$X_L = j\omega L = j800\ \Omega$

$$X_C = \frac{-j}{\omega C} = -j800\ \Omega$$

$$I_0\ (\text{at resonance}) = \frac{V}{R_{net}} = \frac{30}{60}\ A = 0.5\ A$$

so,

a) voltage across $20\,\Omega$ resistor is $= 0.5 \times 20 = 10\,V$

voltage across capacitor $= I_0 X_c = 400\angle -90^\circ\ V$

voltage across coil $= I_0(40 + j800) = 400.5\angle 87.14^\circ\ V$

(b) current $= \dfrac{V}{R + jX_{net}} = \dfrac{30}{60 + j0} = 0.5\ A$

(c) $P = I^2 R_{net} = (0.5)^2 (40 + 20) = 15\ W$

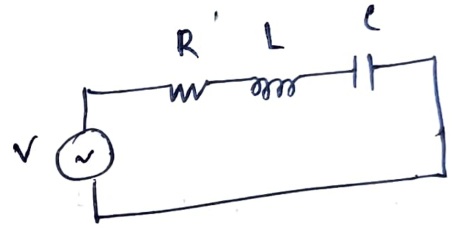
(3) Given

$R = 10\,\Omega$

$L = 30\,mH$

$C = 1\,\mu F$

$V = 10\,V$



Frequency at which voltage developed across capacitor is max^m can be found by $\dfrac{dV_c}{df} = 0$

so,

$$\boxed{f_c = f_0 \sqrt{1 - \frac{R^2 C}{2L}}} \quad \text{— (1)}$$

where $f_c \rightarrow$ frequency at which V_c is max^m

$f_0 \rightarrow$ resonant frequency

so, $f_0 = \dfrac{1}{2\pi\sqrt{LC}} = 918.88\ Hz$

(1) $\Rightarrow$ $\boxed{f_c = 918.11\ Hz}$

so, at f_c

$$X_L = j \times 2\pi \times f_c \times L = j\,173.06\ \Omega$$

$$X_C = \frac{-j}{2\pi \times f_c \times C} = -j\,173.35\ \Omega$$

$$I \text{ at } f_c = \frac{10}{\sqrt{R^2 + (X_{net})^2}} = 1 \angle 1.67\ A$$

so, $\boxed{V_C = I X_C = 173.35 \angle -88.33^\circ\ V}$

(4) dial setting 80 means 800 kHz

so, $f_0 = 800\ kHz$

$$f_0 = \frac{1}{2\pi\sqrt{LC}}$$

$$L = \frac{1}{4\pi^2 C \times f_0^2} \qquad [\text{Put } C = 1\,nF]$$

$$\boxed{L = 39.6\ \mu F}$$

(5) $R = 10\Omega$, $L = 159\,\mu H$, $V = 50mV$, $f_{supplied} = 1MHz$

(a) $C = \frac{1}{4\pi^2 \times L \times f_0^2}$ (at resonance)

take $f_0 = f_{supplied}$

so, $\boxed{C = 159.3\,pF}$

(b) current at resonance $(I_0) = \frac{V}{R} = \frac{0.05}{10} = 5mA$

Now, A/q,

$$0.1 \times I_0 = \frac{5}{\sqrt{10^2 + X_{net}^2}}$$

$$0.5 = \frac{50}{\sqrt{10^2 + X_{net}^2}}$$ (where $X_{net} = X_L - X_C$)

so, $\cancel{X_{net} = 0}$

$\cancel{X_L - X_C = 0}$

$\cancel{X_L = X_C}$

$\cancel{\omega L = \frac{1}{\omega C}}$

$\cancel{C = \frac{1}{\omega^2 L}}$

so, $0.5 = \frac{50}{\sqrt{10^2 + (X_L - X_C)^2}}$

$X_L - X_C = 99.5$ — (i)

$X_L = j \times 2\pi \times f_0 \times 159 \times 10^{-6} = 999.03\Omega$

& $X_C = \frac{-j}{2\pi \times f_0 \times C}$ — (ii)

so, from (i) & (ii) $\Rightarrow$ $\boxed{C = 176.93\,pF}$

⑥ Given $R = 5\Omega$, $L = 4mH$, $C = 0.1\mu F$

max^m rms current flows when circuit is at resonant condition

$$\boxed{I_{0,rms(max)} = \frac{V_{rms}}{R} = \frac{10}{5} = 2\,A}$$

$$\boxed{f_0 = \frac{1}{2\pi\sqrt{LC}} = 7957.74\ Hz}$$

For a series RLC circuit at resonant $cond^n$

$$Q = \frac{1}{R}\sqrt{L/C} = 40$$

$$\boxed{|V_C| = |V_L| = QV = 400\,V}$$

⑦ Given series RLC ckt at resonant $cond^n$.

$$C = \frac{1}{4\pi^2 \times L \times f_0^2}$$

$\left[\ f_0 = 800\,kHz,\ L = 40\mu H,\ R = 4.02\Omega\ \right]$

$$\boxed{C = 989.46\ pF}$$

$$\boxed{B.W. = \frac{R}{L} = 100.5\ rad/s}$$

8. (a) $Q = \frac{1}{R}\sqrt{\frac{L}{C}}$ $R = 4\Omega,\ L = 25\,mH,\ Q = 50$

so, $\boxed{C = 0.625\,\mu F}$

(b) $\omega_1, \omega_2 = \mp\frac{R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 + \frac{1}{LC}}$

so, $\boxed{\omega_1, \omega_2 = 7920\ rad/s,\ 8080\ rad/s}$ respectively

$\boxed{B.W. = \omega_2 - \omega_1 = \frac{R}{L} = 160\ rad/s}$

(c) at ω_r, current flowing in series RLC ckt is $\frac{V}{R}$

so, I_0 at $\omega_r = \frac{100}{4} = 25\ A$

so, $\boxed{P_{avg} \text{ at } \omega_r = I_0^2 \times R = 25^2 \times 4 = 2500\ W}$

we know that at cut-off frequencies i.e. at ω_1 & ω_2, P_{avg} is half of P_{avg} is at ω_r.

so, $\boxed{P_{avg} \text{ at } \omega_1 \text{ \& } \omega_2 = \frac{P_{avg} \text{ at } \omega_r}{2} = 1250\ W}$

(9) The tuner circuit is parallel RLC circuit. When the circuit operates at resonant condⁿ, it catches the selected frequency.

so,

$$C = \frac{1}{4\pi^2 \times L \times f_0^2} \quad \text{——} \quad (1)$$

considering high end of FM band, $f_0 = 108$ MHz

$L = 4 \mu H$ (Given)

so,

(1) $\Rightarrow C = 0.543$ pF

considering low end of FM band, $f_0 = 88$ MHz

(1) $\Rightarrow C = 0.818$ pF

so, range of variable capacitor to cover the entire band is from 0.543 pF to 0.818 pF.